The Impact of Large Language Models (LLMs) in the Health Sector

1. Introduction

Large Language Models (LLMs) are among the most revolutionary advances in artificial intelligence (AI). Constructed using transformer architectures and trained on large datasets, domain-specific models like Med-PaLM and LLMs like GPT-3 and GPT-4 can now produce coherent text, summarise intricate information, and participate in context-sensitive discourse (Maity and Saikia, 2025). Their integration into the health sector has been rapid and wide-ranging, from supporting clinicians in diagnostic decision-making to assisting patients with personalised communication (Yu et al., 2025).

Healthcare represents a particularly high-stakes domain for LLMs because of its reliance on accurate, timely, and ethically informed decision-making. The sector produces vast amounts of unstructured data, clinical notes, research articles, and patient communications, which LLMs are uniquely positioned to process. This makes the evaluation of their potential applications both timely and necessary. However, questions remain about the reliability, transparency, and fairness of these models, especially in contexts where errors could have life-threatening consequences (Benjamens et al., 2020).

This review critically evaluates existing literature on the impact of LLMs in healthcare. It addresses three interrelated dimensions: (1) their clinical applications and outcomes; (2) technological evaluation, including accuracy and limitations; and (3) policy implications concerning safety, regulation, and equity. The target audience for this review includes clinicians seeking practical use cases, developers seeking technical insights, and policymakers

considering governance frameworks. By synthesising current knowledge, highlighting contrasting findings, and identifying gaps, this review aims to contribute an evidence-based perspective on the safe and effective deployment of LLMs in the health sector.

2. Rationale & Significance

The healthcare sector stands at a critical juncture where digital transformation and clinical innovation are converging. LLMs promise to revolutionise how information is accessed, processed, and applied, marking a shift from earlier rule-based systems to models with scale, versatility, and generative capacity (Thirunavukarasu et al., 2023).

Their significance is underscored by growing policy attention: initiatives such as the EU's AI Act and the FDA's evolving frameworks for adaptive AI devices all highlight both the opportunities and risks of deploying LLMs in medicine (European Parliament, 2023; Benjamens et al., 2020).

A clear research gap remains. While early studies and pilot projects show promise, few provide a systematic synthesis of LLM performance, ethical implications, and regulatory readiness. This review, therefore, aims to deliver a rigorous, evidence-based assessment that not only describes LLM innovations but also interrogates their implications for safe, equitable, and sustainable adoption in healthcare.

3. Methodology

Conducting a literature review on the impact of LLMs in healthcare requires a transparent and rigorous methodology to ensure validity. This review employs a narrative synthesis approach, which enables the integration of diverse study designs while highlighting similarities, contrasts, and gaps.

Scope & Search Strategy:

The search focused on publications between 2019 and 2025, reflecting the period in which LLMs gained significant traction in healthcare following the release of models such as GPT-2 and GPT-3. Databases searched included PubMed, IEEE Xplore, ACM Digital Library, Scopus, and Google Scholar. Search terms combined general and domain-specific phrases: "large language models," "transformers," "GPT-3," "GPT-4," "Med-PaLM," "BioGPT," "healthcare," "clinical decision support," "patient communication," and "medical education." Boolean operators (AND/OR) were applied to refine results.

Inclusion/Exclusion Criteria:

- **Included:** Peer-reviewed journal articles, systematic reviews, and preprints that reported empirical findings or conceptual frameworks for LLM applications in healthcare. Both technical evaluations (e.g., performance metrics) and applied studies (e.g., pilot deployments) were included.
- **Excluded:** Non-English papers, purely speculative commentaries without methodological detail, and non-peer-reviewed blogs or opinion pieces.

Data Extraction & Synthesis:

From each included source, details were extracted on (1) model type, (2) healthcare application area, (3) study design, (4) reported outcomes, and (5) identified limitations. Thematic coding was then applied to group findings into categories reflecting the review's structure: evolution of LLMs, healthcare applications, benefits, challenges, and future directions.

By clearly defining scope, criteria, and synthesis strategy, this methodology ensures that the review not only collates available evidence but also systematically highlights areas of

consensus, contention, and uncertainty, providing a foundation for critical analysis in subsequent sections.

4. Overview of LLM Development

The development of large language models (LLMs) has been enabled by advances in neural architectures, data, and computing power. A key breakthrough was the transformer architecture (Vaswani et al., 2017), which became the foundation for GPT-style systems.

Early milestones include **GPT-2**, a scaled-up model that OpenAI initially withheld in full due to concerns about potential misuse, releasing a smaller version instead (Shibuya, 2022). **GPT-3**, with 175 billion parameters, then demonstrated state-of-the-art few-shot learning and set a new scale for performance (Brown et al., 2020). To address the limits of general-purpose models, domain-specific systems such as **BioGPT** (Luo et al., 2022) and **Med-PaLM** (Singhal et al., 2023) were introduced, showing how fine-tuning enhances accuracy in specialised medical contexts.

Training paradigms have also evolved. Reinforcement learning from human feedback (RLHF) improves reliability, while retrieval-augmented generation (RAG) grounds outputs in external knowledge (Ouyang et al., 2022).

This trajectory illustrates both the power and limits of scale: LLMs achieve dramatic gains, but safe healthcare use requires alignment, domain adaptation, and hybrid approaches.

5. Applications of LLMs in Healthcare

The use of large language models (LLMs) in healthcare has expanded since 2020, spanning clinical decision support, research, patient communication, and domain-specific innovations.

While promising, most evidence highlights a gap between experimental results and routine deployment.

5.1 Clinical Decision Support

LLMs are being trialled for diagnosis, triage, and treatment recommendations. GPT-4 has shown accuracy comparable to junior doctors, though performance varies across specialities (Kung et al., 2023). Domain-tuned systems such as Med-PaLM 2 perform well on exam-style questions but remain unproven in clinical environments (Singhal et al., 2023). A recurring concern is hallucination—plausible but incorrect outputs—making oversight essential (Yu et al., 2025; Sallam, 2023).

5.2 Medical Research & Data Analysis

LLMs accelerate research by assisting with literature reviews, hypothesis generation, and data mining. BioGPT (Luo et al., 2022) extracts insights from biomedical texts, while tools like Elicit automate systematic review tasks. However, reproducibility issues persist, with outputs sensitive to prompts and model versions (Thirunavukarasu et al., 2023). Thus, LLMs are best regarded as complements rather than substitutes for human appraisal.

5.3 Patient Communication & Virtual Assistance

Chatbots powered by GPT-4 support triage, scheduling, and health information delivery. They reduce workload and can improve patient satisfaction, especially when responses are empathetic (Nori et al., 2023). Yet risks of misinformation and liability remain, requiring careful governance.

Critical Observations

Across applications, two themes recur: gaps between benchmark performance and clinical reliability, and deployment challenges such as workflow integration, clinician trust, and liability. While results are promising, evidence of safe large-scale implementation is still limited.

6. Benefits & Opportunities

Large language models (LLMs) in healthcare offer benefits that align with both clinical and administrative priorities.

Efficiency and Accessibility: LLMs automate tasks such as drafting notes, summarising histories, and triaging patients. By reducing administrative workload, they free clinicians for direct care, a major advantage amid workforce shortages (Nori et al., 2023). Conversational agents also improve accessibility by delivering health information in plain language, supporting patient engagement.

Accuracy and Safety: Domain-specific systems like Med-PaLM show potential for improved diagnostic accuracy in specialised tasks such as radiology reporting (Singhal et al., 2023). By synthesising complex datasets rapidly, LLMs may reduce oversight risks, though translation of benchmark results into practice still requires validation.

Innovation and Research: Models such as BioGPT mine biomedical literature to generate hypotheses and support drug discovery, accelerating cross-disciplinary research (Luo et al., 2022).

Economic Impact: Automating communication and documentation can reduce labour costs, while efficiency gains may lower hospital stays and readmissions. These savings strengthen the case for adoption, combining clinical and financial incentives.

7. Limitations & Challenges

Despite their promise, large language models (LLMs) face critical limitations that must be addressed before safe, widespread adoption in healthcare.

Reliability and Transparency: A central risk is hallucination—plausible but incorrect outputs that are dangerous in clinical contexts (Thirunavukarasu et al., 2023). Although reinforcement learning from human feedback (RLHF) can reduce errors, it does not eliminate them. Compounding this, LLMs remain "black boxes," with opaque reasoning that raises accountability concerns (Amann et al., 2020).

Privacy, Bias, and Equity: Training often involves sensitive datasets, creating risks of privacy breaches. Without careful curation, models may reproduce existing biases, disproportionately affecting underrepresented populations (Vyas et al., 2020). Addressing these issues requires inclusive datasets and fairness-aware modelling.

Evaluation and Regulation: Impressive benchmark scores, such as those from Med-PaLM, do not yet equate to clinical reliability. Standardised evaluation metrics and clinical trial-style validations are scarce, leaving regulators and clinicians without robust guidance. Early frameworks like the FDA's adaptive AI guidance and the EU AI Act provide direction but remain incomplete (European Parliament, 2023).

Sustainability: LLMs demand substantial computational resources, raising environmental concerns and limiting access for resource-constrained health systems.

Taken together, these challenges highlight the need for rigorous evaluation, transparency, and governance. Without them, LLMs risk undermining trust and amplifying inequities rather than improving healthcare delivery.

8. Thematic Comparison & Critical Analysis

A distinguishing feature of this review is its commitment to critical synthesis rather than simple description. The literature indicates both convergence and divergence in how LLMs are perceived across application areas. To illustrate, models such as GPT‑4 and Med‑PaLM show strong performance in benchmark testing, yet this does not consistently translate into reliable deployment in clinical practice (Singhal et al., 2023; Nori et al., 2023). Similarly, while BioGPT demonstrates the potential of domain-specific adaptation for biomedical research, concerns about reproducibility and validation persist (Luo et al., 2022).

A useful way to evaluate these tensions is through comparative mapping. For example, clinical decision support promises efficiency but raises safety concerns, whereas patient communication tools enhance accessibility but risk misinformation. Cross-comparison highlights a recurring theme: trade‑offs between efficiency and safety, innovation and reliability, accessibility and equity.

Critically, the review also highlights methodological limitations across the literature. Many studies are proof-of-concept trials or rely on exam-style benchmarks rather than longitudinal patient outcomes. Few works provide robust, standardised metrics, leaving policymakers and clinicians without clear evidence to guide adoption. This comparative lens shows that while enthusiasm is high, evidence for safe, equitable integration remains fragmented and uneven.

9. Future Directions

The future trajectory of LLMs in healthcare will depend on addressing current limitations while capitalising on their strengths. Several avenues emerge from the literature:

Standardisation and Evaluation Frameworks: There is an urgent need for shared benchmarks and clinical trial-style validations. Without these, claims of "expert-level" performance remain difficult to substantiate (European Parliament, 2023).

Multimodality and Domain Adaptation: Integrating multimodal inputs, such as text with imaging or genomic data, could improve diagnostic accuracy. Domain-specific models like Med‑PaLM exemplify how adaptation enhances safety and relevance (Singhal et al., 2023).

Bias Mitigation and Equity: Future research must prioritise inclusive datasets and fairness-aware algorithms. As Vyas et al. (2020) argue, clinical algorithms that neglect demographic diversity risk amplifying health inequities.

Policy and Governance: Regulatory initiatives like the EU AI Act and FDA guidance will play a critical role in shaping safe adoption. These frameworks should evolve in tandem with technological advances to ensure accountability, transparency, and ethical use.

Collaborative Ecosystems: Effective deployment requires collaboration among clinicians, AI developers, ethicists, and policymakers. Interdisciplinary governance can bridge technical innovation with real-world clinical needs.

Overall, the literature suggests a cautious but optimistic outlook: LLMs could transform healthcare if their adoption is guided by rigorous evaluation, fairness, and policy alignment.

10. Conclusion

Large language models (LLMs) represent a transformative shift in healthcare, offering unprecedented capabilities for processing clinical data, supporting decision-making, advancing research, and improving communication. This review has highlighted how their integration spans multiple domains, from clinical decision support and medical research to patient engagement and multimodal innovations, demonstrating both breadth and versatility.

However, evidence also reveals a clear distinction between benchmark performance and real-world deployment. While domain-tuned systems like Med-PaLM have achieved expert-level scores on examinations, their reliability in routine practice is less certain. The risks of hallucinations, opacity, bias, and inequitable access underscore the need for cautious adoption under strong governance.

The rationale for this review was grounded in the urgency of LLM adoption and the policy attention surrounding AI regulation, exemplified by the EU AI Act and FDA initiatives. These frameworks signal recognition of both the opportunities and dangers of applying LLMs in high-stakes clinical contexts.

Ultimately, the future of LLMs in healthcare depends on aligning technological innovation with clinical validation, regulatory oversight, and ethical responsibility. To harness their full potential, stakeholders must address gaps in evaluation, transparency, and equity. Only then can LLMs be responsibly deployed to deliver safer, more efficient, and more equitable healthcare systems worldwide.

References

Amann, J., Blasimme, A., Vayena, E., Frey, D. & Madai, V.I. (2020) 'Explainability for artificial intelligence in healthcare: a multidisciplinary perspective', *BMC Medical Informatics and Decision Making*, 20(1), 310. Available at: https://doi.org/10.1186/s12911-020-01332-6 (Accessed: 10 September 2025).

Benjamens, S., Dhunnoo, P. & Meskó, B. (2020) 'The state of artificial intelligence-based FDA-approved medical devices and algorithms: an online database', npj *Digital Medicine*, 3, 118. Available at: https://doi.org/10.1038/s41746-020-00324-0 (Accessed: 10 September 2025).

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., … & Amodei, D. (2020) 'Language models are few-shot learners', *Advances in Neural Information Processing Systems*, 33, pp. 1877–1901. Available at: https://arxiv.org/abs/2005.14165 (Accessed: 10 September 2025).

European Parliament (2023) 'EU AI Act: first regulation on artificial intelligence'. Available at: https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence#:~:text=AI%20regulation%20in%20Europe%3A%20the,first%20comprehensive%20framework (Accessed: 10 September 2025).

Kung, T.H., Cheatham, M., Medenilla, A., Sillos, C., De Leon, L., Elepaño, C., Madriaga, M., Aggabao, R., Diaz-Candido, G., & Maningo, J. (2023) 'Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models', *PLOS Digital Health*, 2(2), e0000198. Available at: https://doi.org/10.1371/journal.pdig.0000198 (Accessed: 10 September 2025).

Luo, R., Sun, L., Xia, Y., Qin, T., Zhang, S., Poon, H. & Liu, T.Y. (2022) 'BioGPT: generative pre-trained transformer for biomedical text generation and mining', *Briefings in Bioinformatics*, 23(6), bbac409. Available at: https://doi.org/10.1093/bib/bbac409 (Accessed: 10 September 2025).

Maity, S. & Saikia, M.J. (2025) 'Large language models in healthcare and medical applications: a review', *Bioengineering (Basel)*, 12(6), p. 631. Available at: https://www.mdpi.com/2306-5354/12/6/631 (Accessed: 10 September 2025).

Nori, H., King, N., McKinney, S.M. & Carignan, D. (2023) 'Capabilities of GPT-4 on medical challenge problems', *arXiv preprint*. arXiv:2303.13375. Available at: https://arxiv.org/abs/2303.13375 (Accessed: 10 September 2025).

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., … & Christiano, P. (2022) 'Training language models to follow instructions with human feedback', *arXiv preprint*. arXiv:2203.02155. Available at: https://arxiv.org/abs/2203.02155 (Accessed: 10 September 2025).

Sallam, M. (2023) 'ChatGPT Utility in Healthcare Education, Research, and Practice: Systematic Review on the Promising Perspectives and Valid Concerns', *Healthcare*, 11(6), 887. Available at: https://doi.org/10.3390/healthcare11060887 (Accessed: 10 September 2025).

Shibuya, N. (2022). *GPT-2: Too Dangerous to Release (2019)*. Naoki Shibuya Blog. Available at: https://naokishibuya.medium.com/gpt-2-too-dangerous-to-release-2019-4a0b0ff7db09 (Accessed: 10 September 2025).

Singhal, K., Tu, T., Gottweis, J., Sayres, R., Wulczyn, E., Hou, L., … & Corrado, G.S. (2023) 'Towards expert-level medical question answering with large language models', *Nature*, 620, pp. 545–552. Available at: https://doi.org/10.1038/s41586-023-06291-2 (Accessed: 10 September 2025).

Thirunavukarasu, A.J., Ting, D.S.J., Elangovan, K. & Nair, R. (2023) 'Large language models in medicine', *Nature Medicine*, 29, pp. 1930–1940. Available at: https://doi.org/10.1038/s41591-023-02448-8 (Accessed: 10 September 2025).

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. & Polosukhin, I. (2017) 'Attention is all you need', *Advances in Neural Information Processing Systems*, 30. Available at: https://papers.nips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html (Accessed: 10 September 2025).

Vyas, D.A., Eisenstein, L.G. & Jones, D.S. (2020) 'Hidden in plain sight - reconsidering the use of race correction in clinical algorithms', *New England Journal of Medicine*, 383(9), pp. 874–882. Available at: https://doi.org/10.1056/NEJMms2004740 (Accessed: 10 September 2025).

Yu, E., Chu, X., Zhang, W., Meng, X., Yang, Y., Ji, X. & Wu, C. (2025) 'Large language models in medicine: applications, challenges, and future directions', *International Journal of Medical Sciences*, 22(11), pp. 2792–2801. Available at: https://www.medsci.org/v22p2792.htm (Accessed: 10 September 2025).